Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, April/May - 2023
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define Early Effect. [2]
b) What is meant by Depletion Region? [3]
c) What are the Harmonic components in a Rectifier? [2]
d) Define Percentage of Regulation. [3]
e) Draw the diode equivalent model of a transistor. [2]
f) Find the value of β if a transistor has $\alpha = 0.97$. Also find α if $\beta=200$. [3]
g) What is Thermal runaway in transistors and mention how it can be avoided? [2]
h) Why a capacitive coupling used to connect signal source to an amplifier. [3]
i) Differentiate the FET operation in enhancement and depletion modes. [2]
j) Why the input impedance of FET is higher than BJT? [3]

PART – B

(50 Marks)

2.a) Draw and explain the graph indicating the variation of minority carrier density with distance in a p-n junction diode under forward biased condition.
b) Derive expression of space charge capacitance C_T for a step graded junction. [5+5]

OR

3.a) Derive the expression for Diffusion Capacitance.
b) Discuss the static characteristics of SCR with a neat diagram. [4+6]

4.a) Derive the expression for ripple factor of Half wave rectifier with C-filter.
b) A full-wave single phase rectifier employs a π- section filter consisting of a pair of 5 μF capacitances and a 20 H choke. The transformer voltage to the center tap is 200 V rms. The load current is 600 mA. Calculate the dc output voltage and the ripple voltage. The resistance of the choke is 300Ω. [6+4]

OR

5.a) Derive V_{dc}, V_{rms} and PIV for the circuit of a Full-wave rectifier and explain its operation.
b) Describe the working of a Zener Diode as a Voltage Regulator. [5+5]

6.a) With a neat diagram, explain the operation of BJT and discuss about the different configurations.

b) Explain the hybrid small signal model for common collector configuration. [5+5]

OR

7.a) With neat diagram, explain the input and output characteristics of a transistor in CE configuration.

b) Describe the operation of UJT as relaxation oscillator. [5+5]

8.a) Explain DC load line and Q point for any transistor configuration. Also state the necessity of biasing and list biasing methods for transistor.

b) Determine the quiescent currents and the collector to emitter voltage for a Ge transistor with $\beta = 50$ in the Self biasing arrangements. The circuit component values are $V_{CC} = 20V$, $R_C = 2k\Omega$, $R_E = 0.1\ k\Omega$, $R_1 = 100\ k\Omega$ and $R_2 = 5\ k\Omega$. Find the stability factor S. [5+5]

OR

9.a) Draw and explain self-bias circuit. Mention its advantages over fixed bias circuit.

b) Describe the analysis of Transistor Amplifier circuit using h-parameters. [5+5]

10.a) Illustrate the construction and operation of p-channel JFET with neat diagrams

b) With neat sketch, discuss the working of a common source FET amplifier. [5+5]

OR

11.a) With neat sketch, describe the working principle, operation and characteristics of the N-channel Enhancement mode MOSFET.

b) Explain how JFET works as a Voltage Variable Resistor? [5+5]

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